Work achievements

April 2024 → January 2025

CAMILLE IBRAHIM

DATA SCIENTIST CONTRACTOR @VPE DRIVEABILITY

Project Goal :

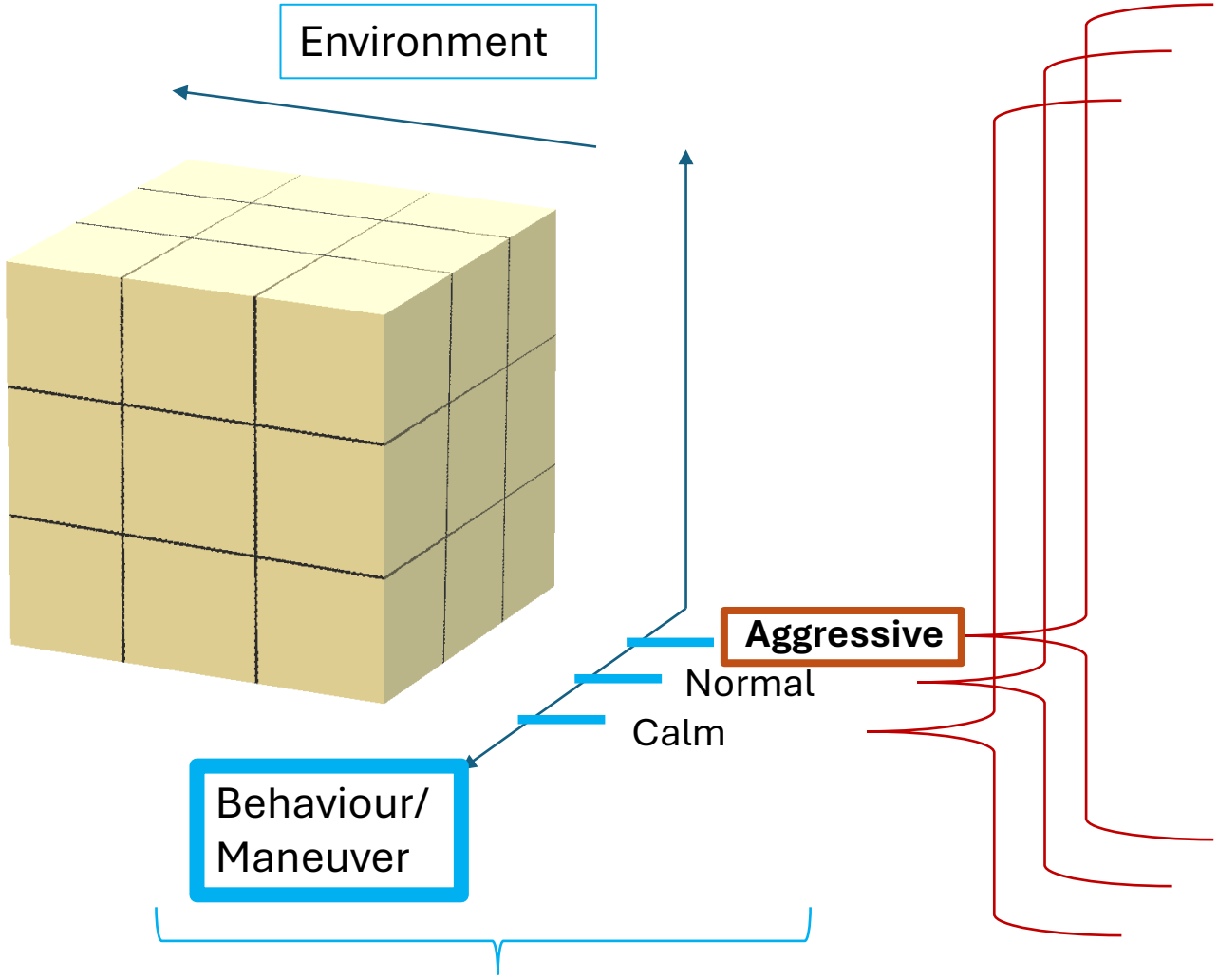
The objective of this project was to evaluate driving behavior, with a focus on aggressiveness, across different driving environments. Each driver is represented by a unique 3D matrix defined by three axes: **Behavior – Environment – Habits**.
My responsibility was to implement a new model that provides insights related to behavior, one of these key axes.
In parallel, I developed several other codes for different purposes. Below are the most relevant ones that showcase my skills.

DATA USED : Time Series telemetry data of all vehicles models available in CEP, acquisition frequency : 100 Hz

1) I implemented scripts that detect different types of maneuvers (acceleration, deceleration, cornering, following a car) using methods implemented by ADAS team. I collaborated with them to have better insight about the logic of the functions they implemented.

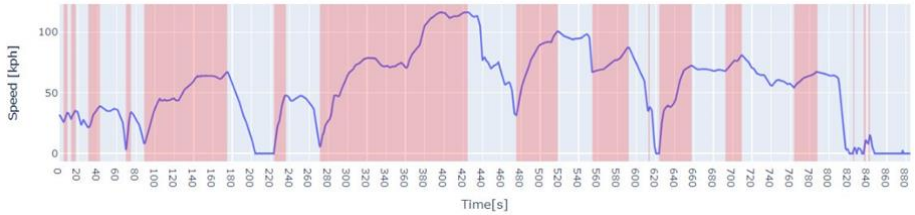
I also checked the quality of the signals and reconsider which signals were actually best to use, since there are discrepancies depending on the car model we consider (quality, availability, units, ...)
I improved the quality of some signals (corrected artifacts of speed signal, modified the smoothiness of acceleration, computed the lateral acceleration) and after discussion with engineers specialized in maneuvers, I improved the detection of cornering maneuvers.

The idea is to represent a driver by its unique 3D matrix

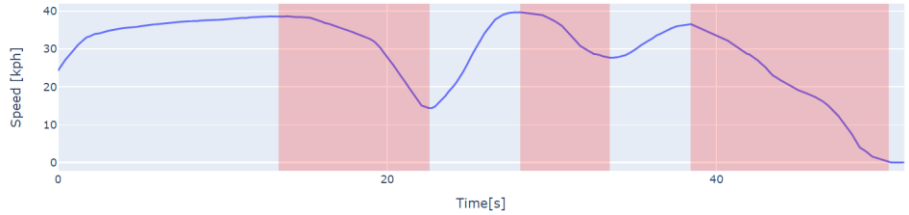


My goal is to evaluate the aggressiveness of the driver's maneuvers

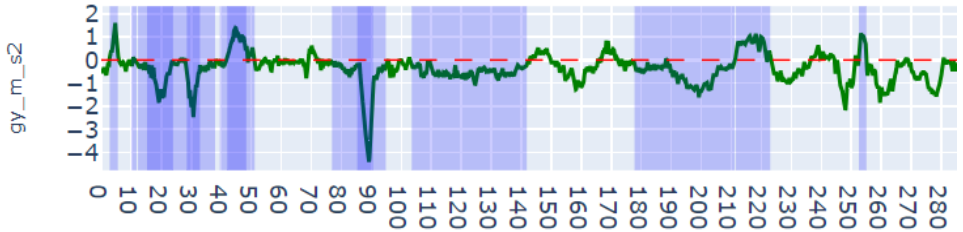
Acceleration



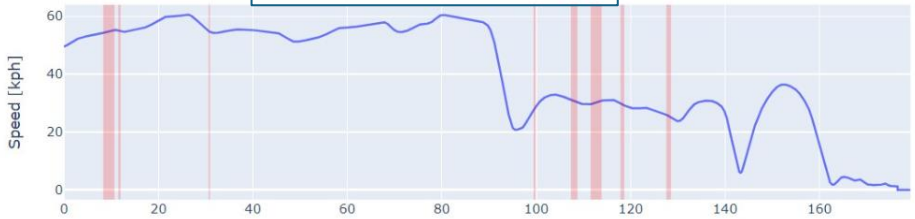
Deceleration



Cornering



Following a car



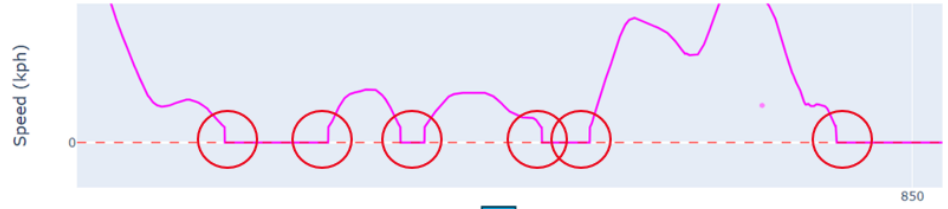
Maneuvers have to be detected first (highlighted)

Cornering detection improved

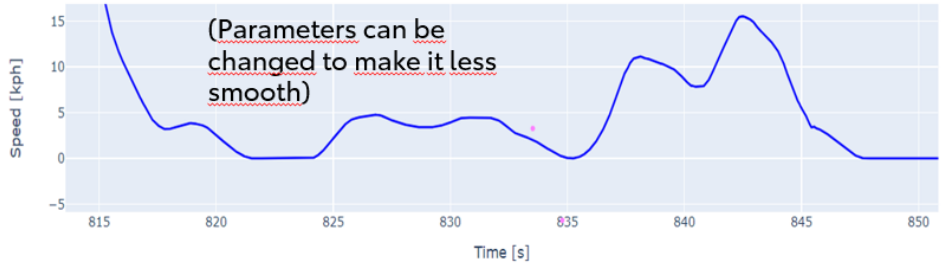
Undetected cornerings **ADAS**

Now detected **After modification**

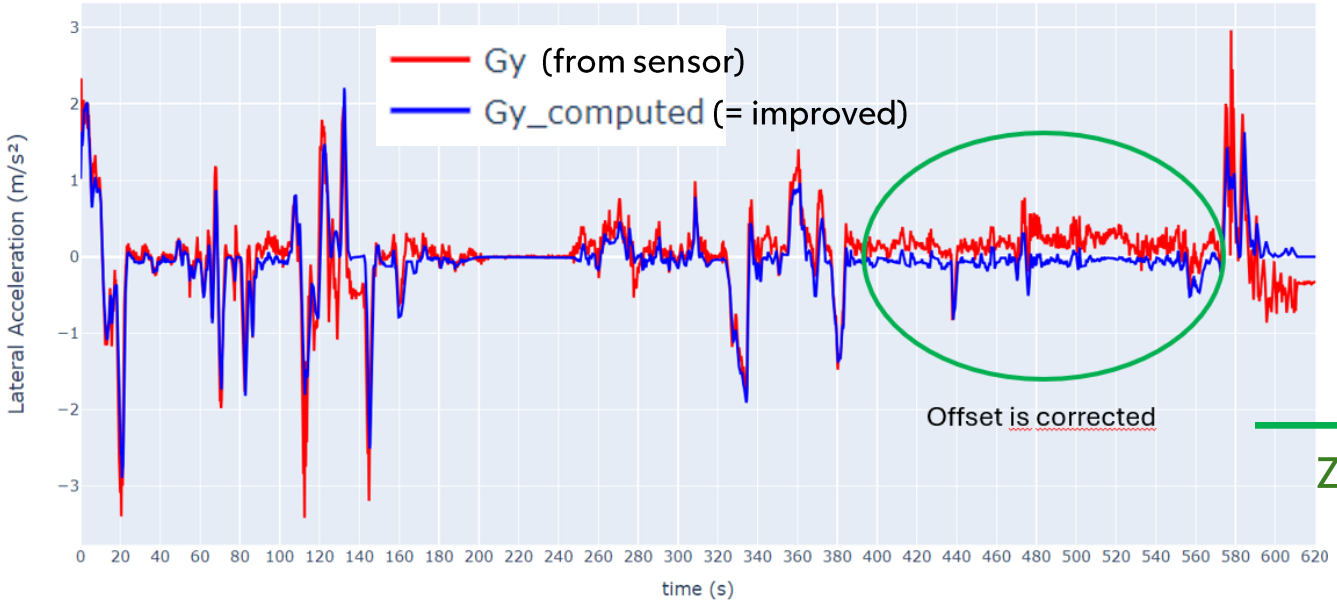
Artifacts (jumps) of Speed signal from sensors corrected



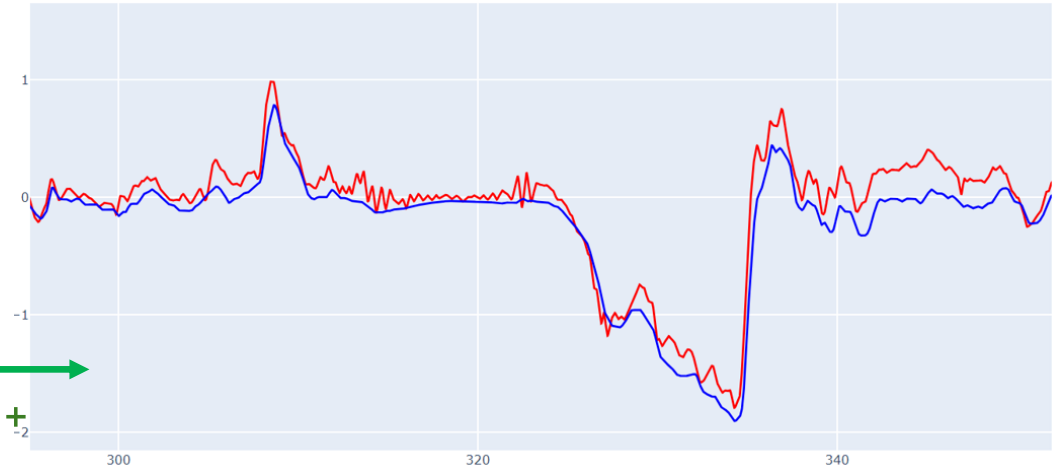
Speed over Time



Lateral acceleration signal quality improved (computed)



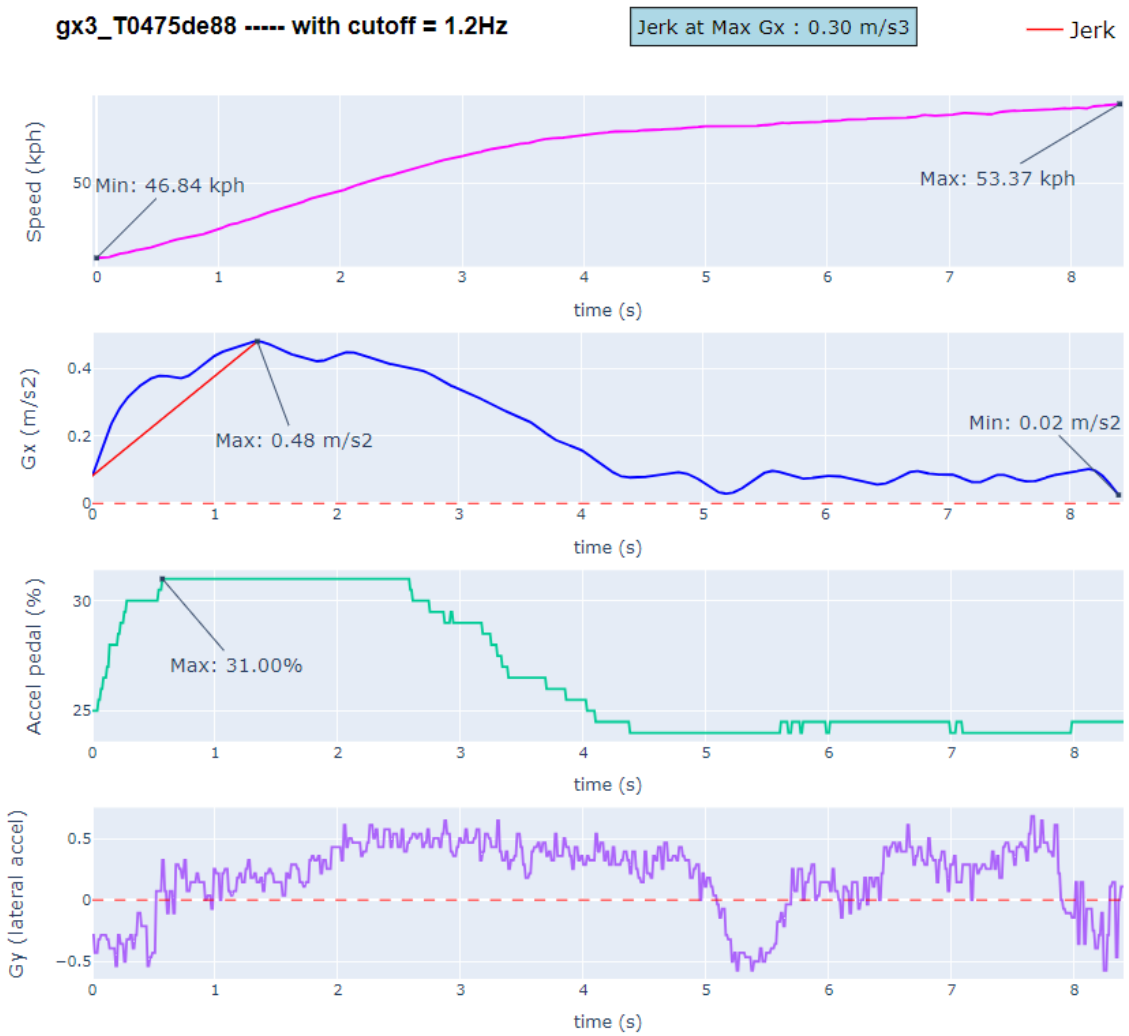
ZOOM+



2) I implemented a new pipeline that computes specific KPIs for every maneuvers. The pipeline is available for engineers who study maneuvers, is also used to feed the T-le Metrics dashboard implemented within our team, and is used for evaluating the aggressiveness that I implemented.
I created a user-friendly notebook that runs this pipeline for any engineer to use it, with almost no need for prior knowledge in Python.

1 Trip

1 Acceleration maneuver



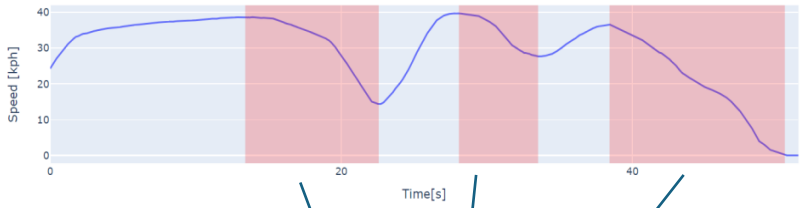
- Acceleration maneuver detected (« X »)
- KPIs computed for that maneuver + plots (right)
- Graph analysis and validation

Retrieve KPIs from [Accel – Decel – Cornering] from ALL the vehicles and save them into files

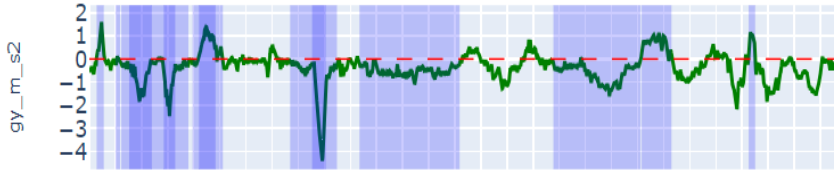
Go through all trip paths and filter out invalid trips. Keep only valid trips. Processes are done trip by trip.

Acceleration

Deceleratio



Cornering



KPIs

SPEED (MAX, MEAN ..)
Gx (MAX, MEAN ..)
JERK
DURATION OF MANEUVER

GPS COORDINATES
ENVIRONMENT
ACC
TARGET

MODEL TYPE
ENGINE TYPE
COUNTRY
PRODUCTION DATE

KPIs

SPEED (MAX, MEAN ..)
Gy (MAX, MEAN ..)
STEERING ANGLE
YAW ANGLE

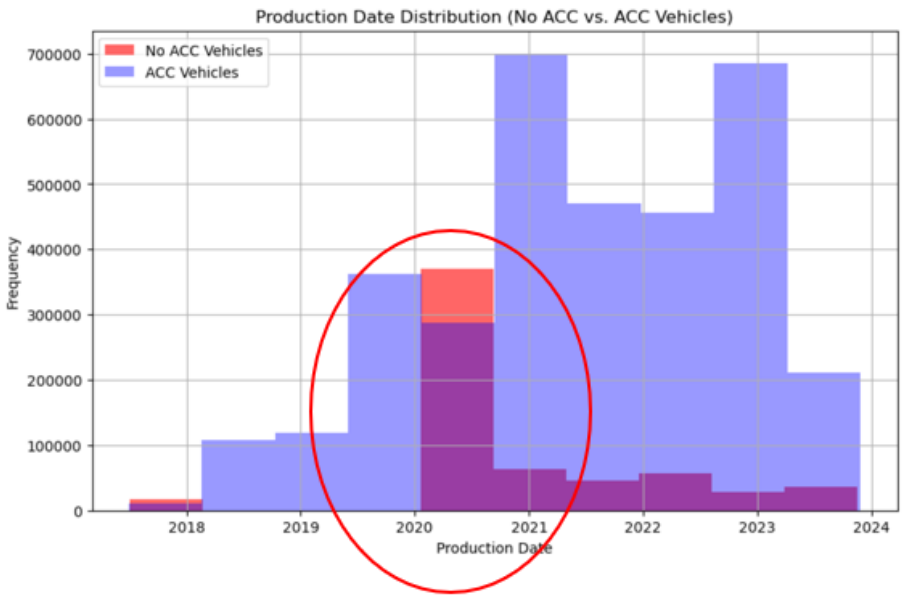
GPS COORDINATES
ENVIRONMENT
ACC
TARGET

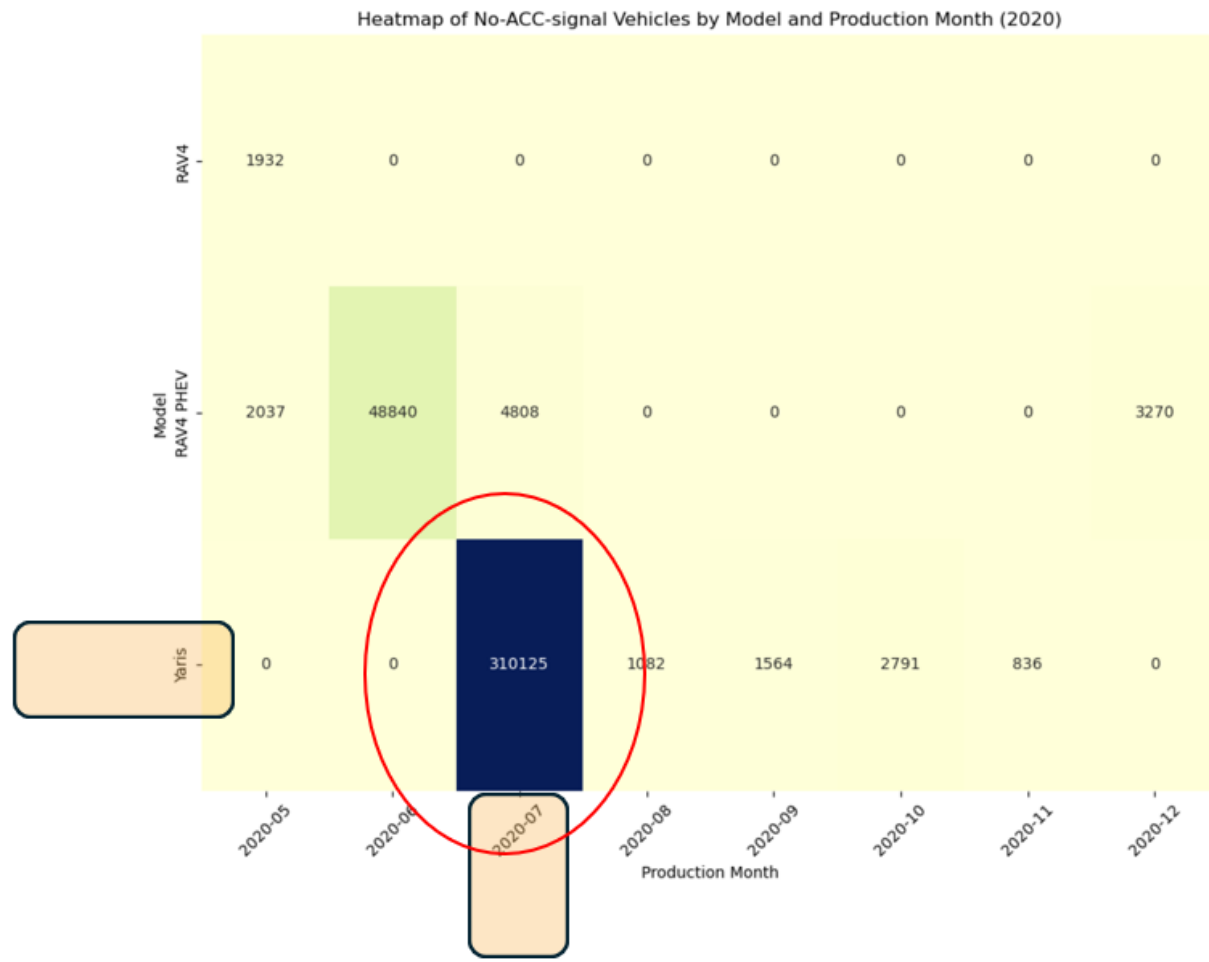
MODEL TYPE
ENGINE TYPE
COUNTRY
PRODUCTION DATE

3) I conducted an in-depth investigation of invalid trips to ensure they were not considered in the aggressiveness model I implemented. This involved plotting histograms and pie charts, verifying the availability of necessary signals, and making decisions on whether to exclude trips or replace missing signals with alternative ones.
For example, the Yaris model lacked information on ACC (Adaptive Cruise Control). Therefore, only trips where ACC was OFF were considered. Another example, no signal for ACC was recorded at all for Aygo X, therefore I had to find a spare signal that has real values recorded only if ACC was enabled. So, in the code, a specific condition needed to be added for Aygo X.

YARIS

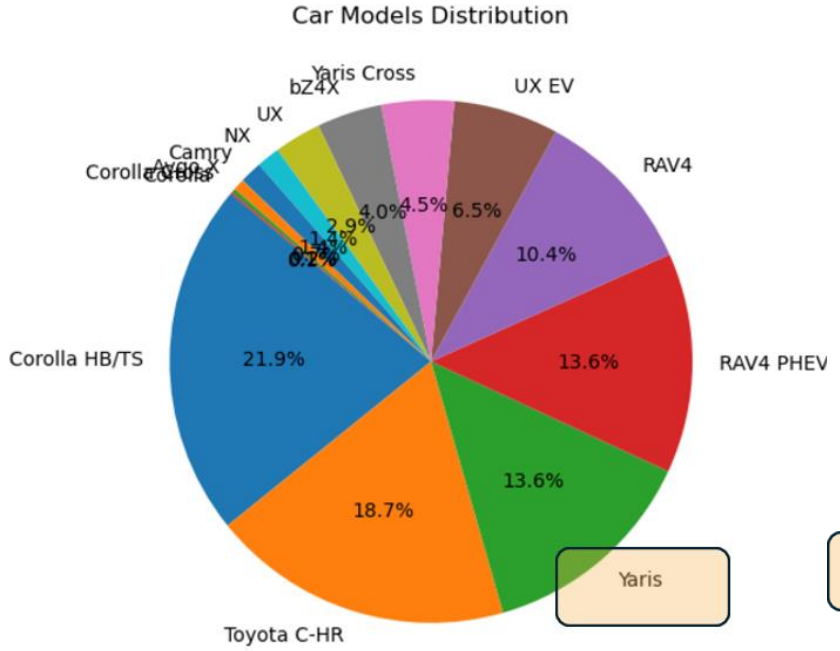
Reason for lack of information about ACC

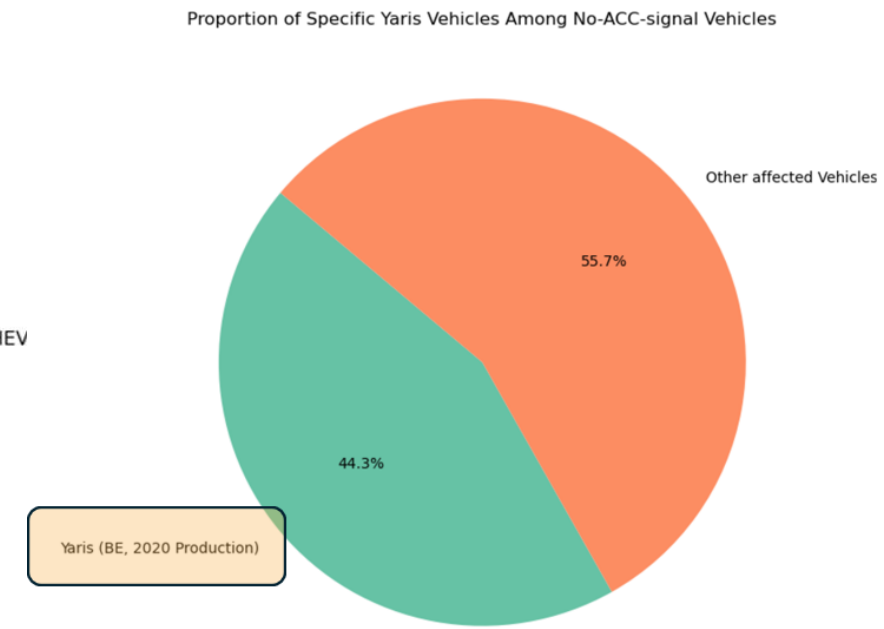


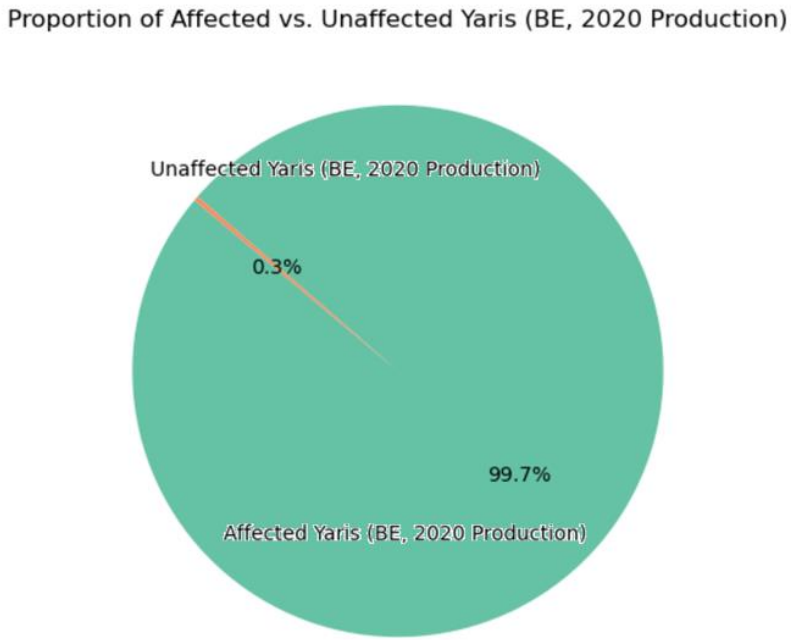


YARIS

Reason for lack of information about ACC







Among all the vehicles without ACC signal, this specific Yaris covers almost half of the reasons why we have these NaNs in the data

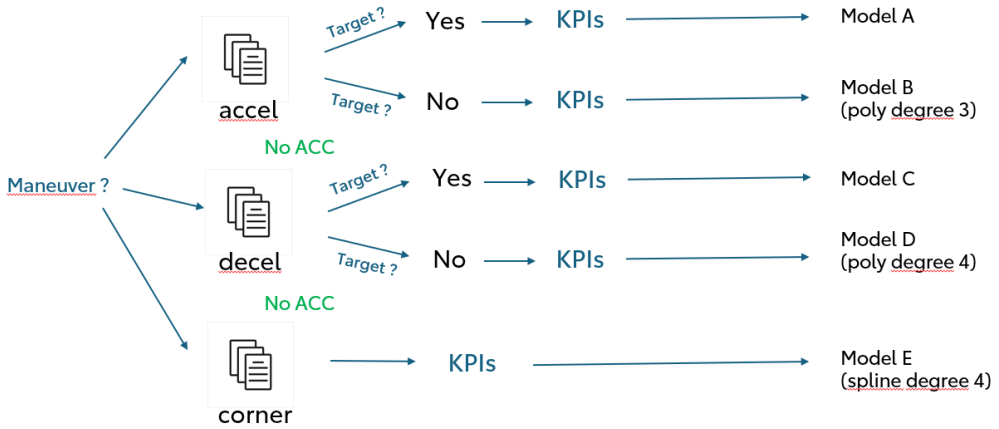
And it concerns all the vehicles that are [Yaris, BE, 2020 production]

Simple reason is it hadn't been recorded

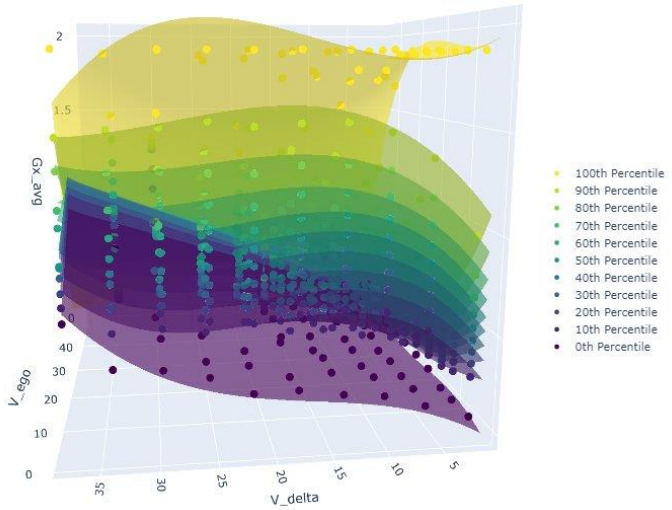
4) I developed a new pipeline that evaluates the aggressiveness of each type of driving maneuver (accel without vehicle in front, decel without vehicle in front, cornering, following a car), based on the driving environment (urban, rural, highway, ..). The final output is a personalized heatmap for each driver displaying the concentration of aggressiveness percentiles for their maneuvers. This is done independently for each maneuver type and adjusted for the driving environment.
It can answer questions like :
→ « How aggressively does this person accelerate when driving in a congested city ? »

I carefully selected the maneuvers to be analyzed, ensuring that cases involving cruise control or a car in front were excluded. Additionally, I conducted statistical checks to verify the number of maneuvers for each car model.
To implement the aggressiveness model, I considered the entire dataset (all vehicles and car models).
I then designed an architecture that follows a logical process:
•Depending on the type of maneuver, it computes the relevant KPIs.
•It selects the appropriate model.
•It generates specific visualizations.
The entire code is structured to run with a single line of input, requiring only the necessary parameters based on the desired analysis :

```
Aggressiveness(car_in_front=False, acceleration=True, deceleration=False, cornering=False,
               percentiles, csv_path, new_point, plot=True, models_exist=True)
```

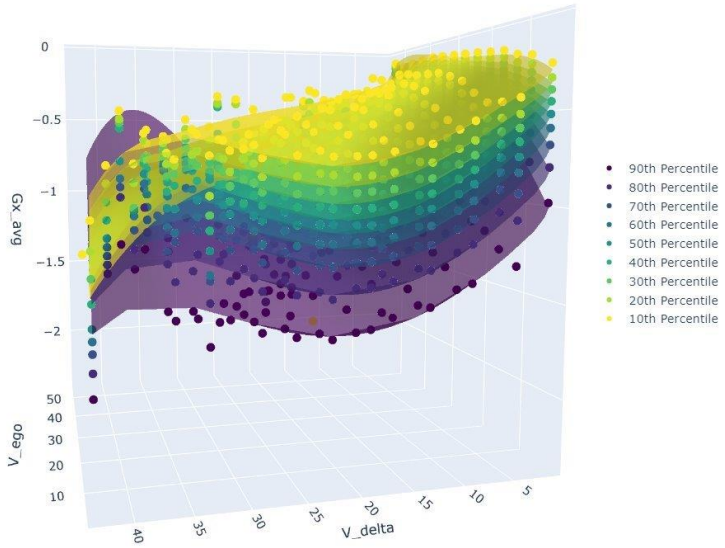


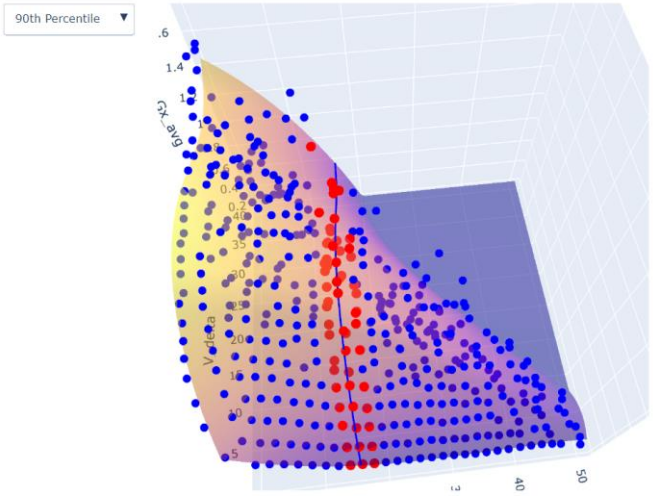
ACCELERATION

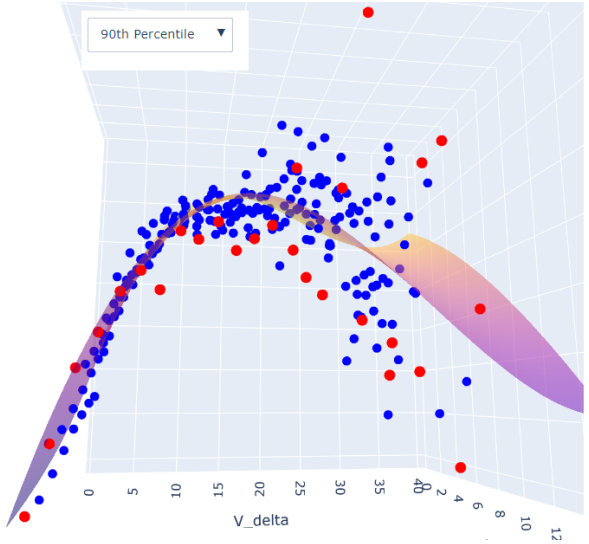


Visualization of the quality of the models' fitting with a 3D rotatable figure where we can select the surface, truncate it to visualize from the side etc.

DECELERATION

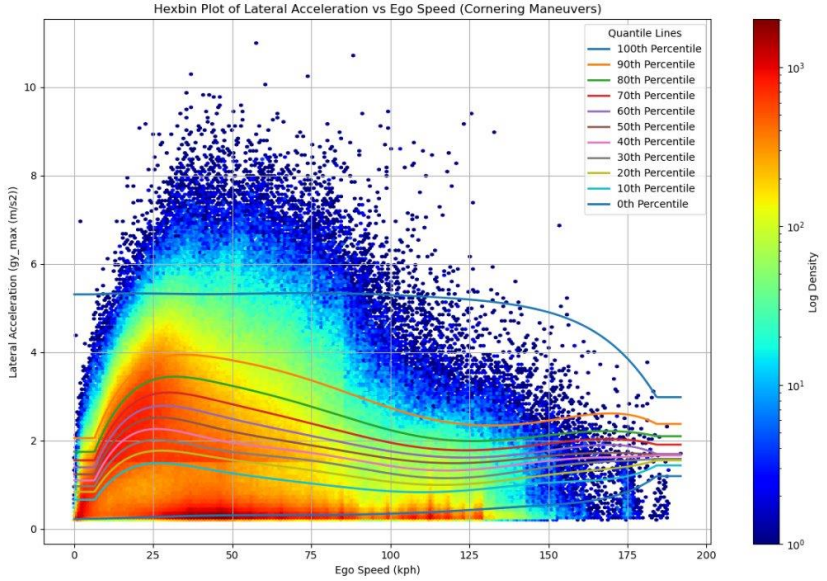


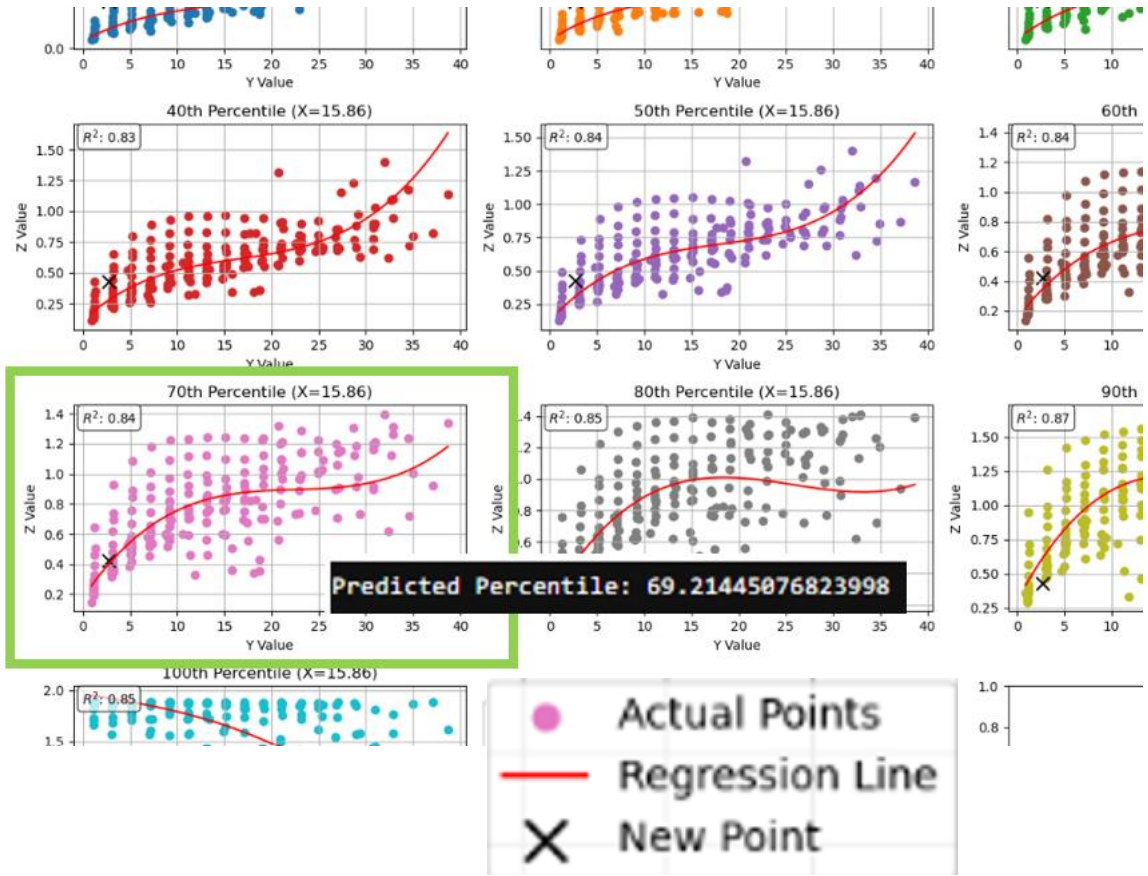


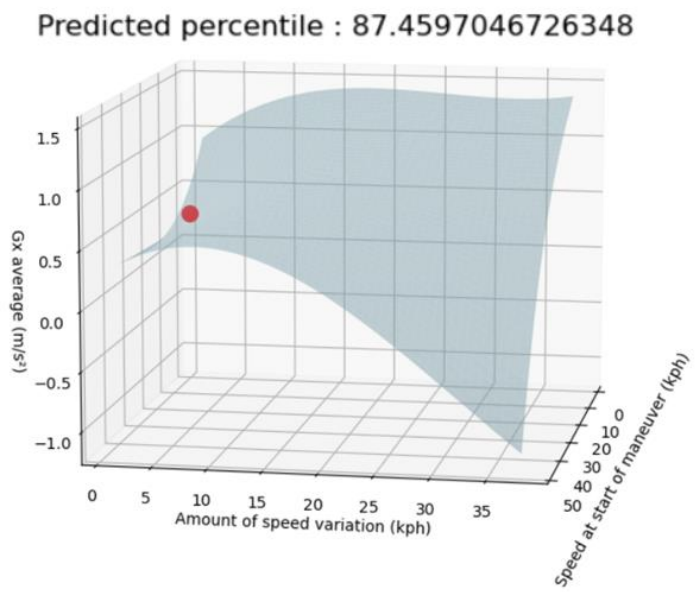


90th percentile regression surface + data points,
view from the top and from the side

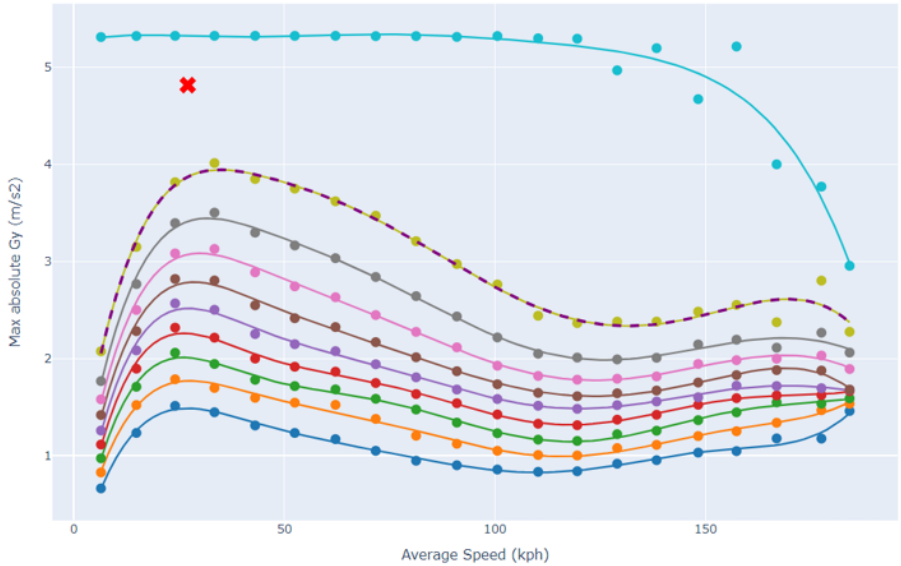
CORNERING







Regression lines and predicted new point for cornering - Predicted percentile : 94



From the model trained, we can thus predict the percentile of aggressiveness for any new data point (one maneuver), and we do it for all the maneuvers of a driver

Example of a heatmap representing the occurrences of aggressive acceleration maneuvers in each percentiles interval and by environment, of a Corolla HB user.

Occurences of aggressive accel maneuvers by environment and percentiles interval for vehicleID 0a2598c1-91b0-4c06-928b-6308e7918fe7

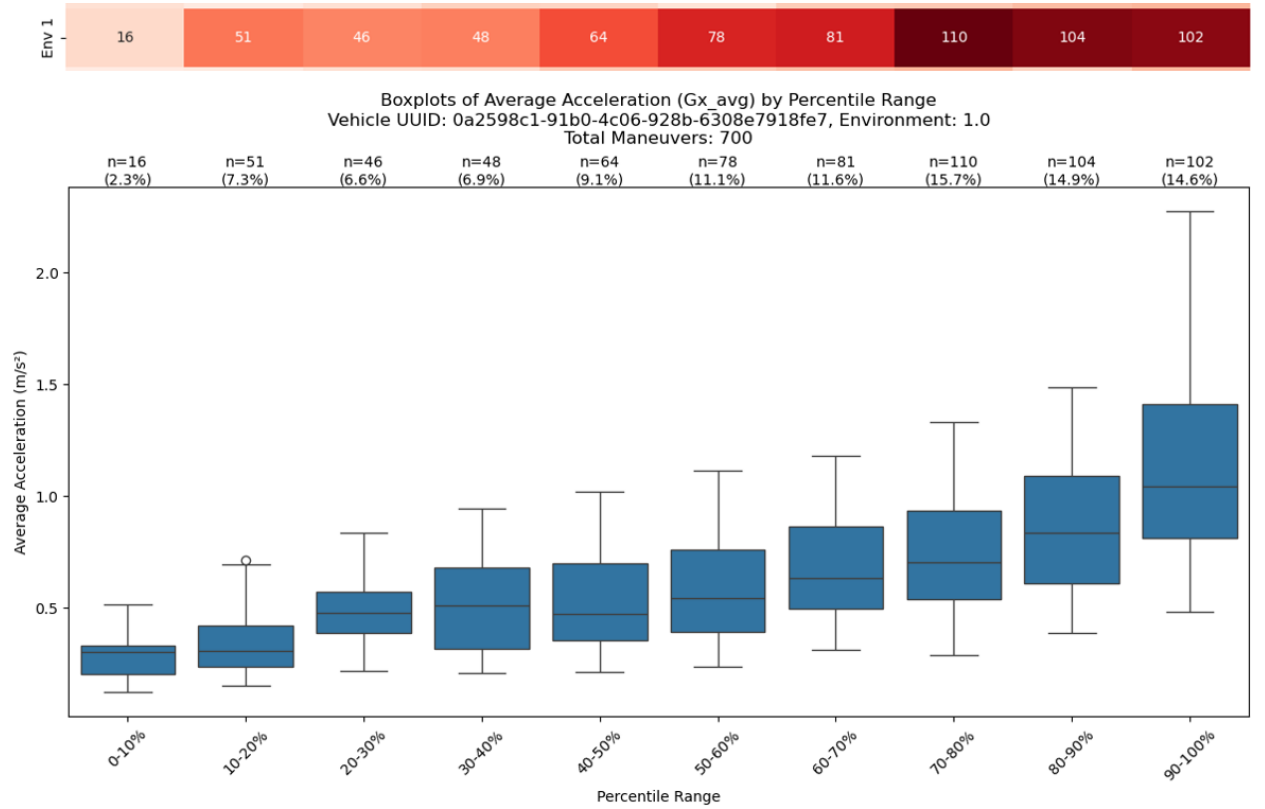
Environments	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90	90-100
Env 0	13	7	13	19	18	11	21	22	27	29
Env 1	16	51	46	48	64	78	81	110	104	102
Env 2	6	8	14	15	22	22	26	29	19	28
Env 3	2	3	4	4	12	3	8	10	6	6
Env 4	5	3	7	7	3	3	5	4	7	7
Env 5	0	2	5	2	4	7	8	14	11	14
Env 6	2	12	9	5	9	8	13	12	7	15
Env 7	0	0	0	0	3	3	2	4	4	2
Env 8	1	0	0	0	0	0	0	2	0	1
Env 9	1	0	1	2	0	4	2	2	3	0
Env 10	0	3	4	4	6	3	3	4	2	1
Env 11	0	2	0	1	1	4	2	0	1	1

Percentiles

Urban – Center Start&Stop

Urban – Center Busy

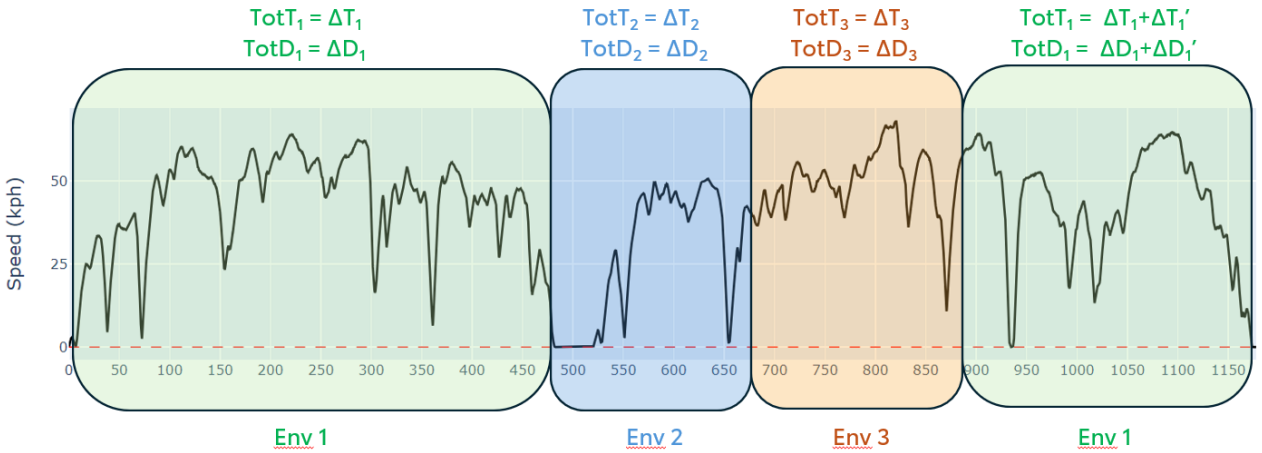
Urban – Center Free flow

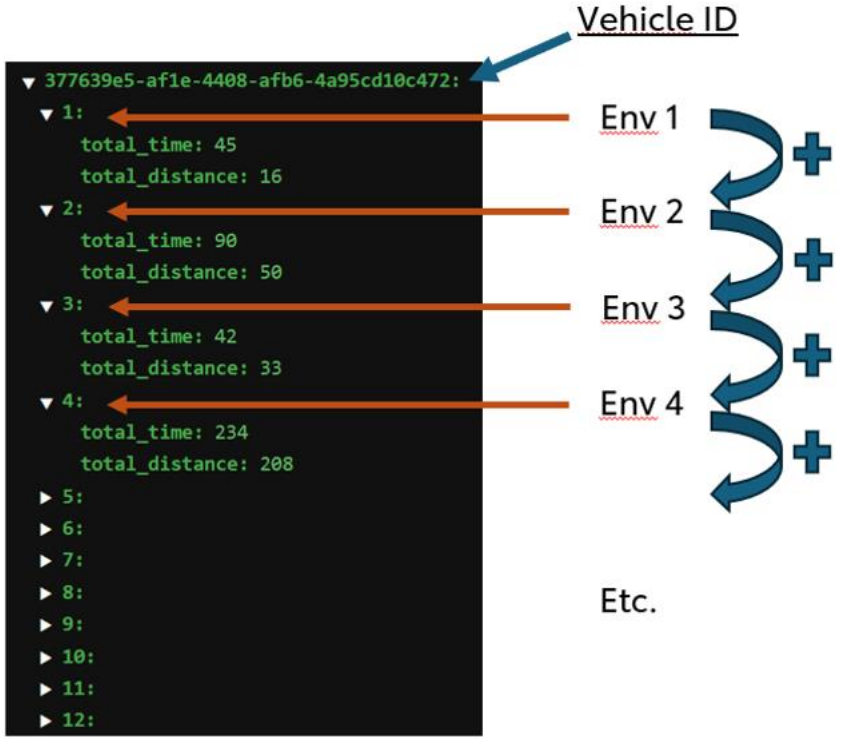


High number of occurrences in high percentiles range
→ Driver is considered aggressive in the specific type of environment

We cannot simply judge and compare vehicles' behaviours only based on the matrices.

→ **Need to consider in parallel Time spent and Distance covered in each environment**



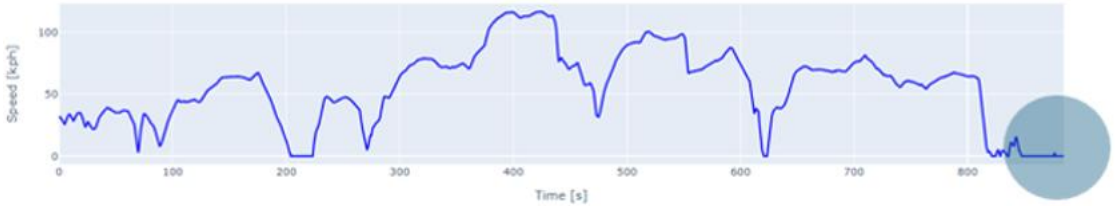


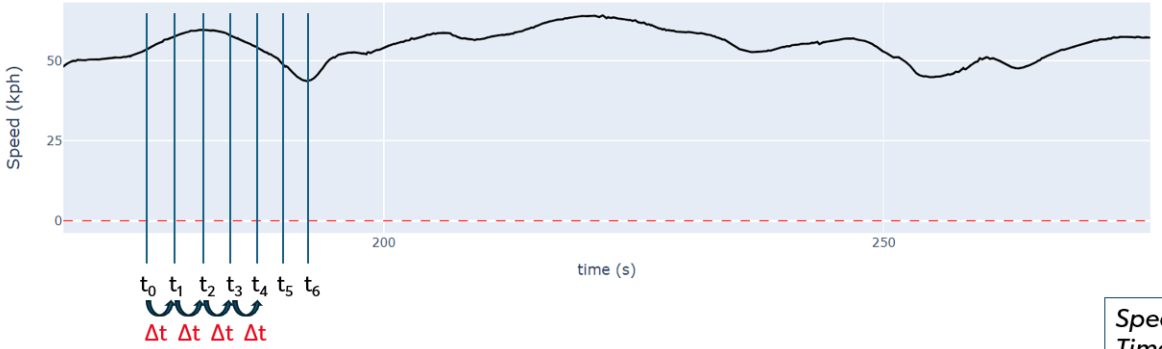
For each vehicle, individually :

(1) From the matrice, I computed :
- Total time covered (all trips, every environments)
- Total distance covered (all trips, every environments)

They need to be aligned. However, discrepancies helped me identify trips with invalid environment labels.

(2) From the raw trips, I computed :
- Sum of all latest timestamps (e.g. duration)
- Distance from time and speed values





(Part of a trip ZOOMED++)

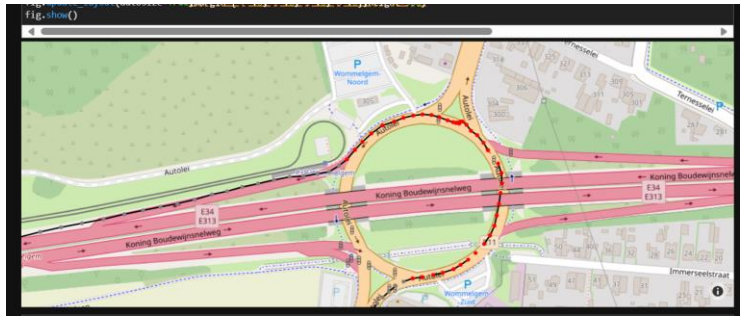
Total distance per trip = $\Sigma\ (\Delta t * v_{t,i})$

Speed → kph
Timestamp → sec
Distance → km
*(D = t * v / 3600)*

API requests and merging data

Latest task ongoing : I collaborated again with ADAS in order to add features related to Road attributes.
I implemented a new pipeline to process the trips of our own database, **make API calls**, and **merge** the new data with our existing data.

Mapping	Routing	Road attributes
Display interactive maps in various styles	Generate routes for driving, walking, etc.	Information about Speed limit, Traffic lights, Stop signs, Roundabouts, Intersections, etc.
Embedding maps into web or mobile apps	Route optimization (choose best route)	
Geocoding / Reverse geocoding Address ← → Coordinates	Traffic information	



To ensure the reusability of the code, I documented how the pipeline works. I applied this documentation process to all the pipelines I created, including details on challenging points and how they were addressed.